

X-ray Photoelectron Spectroscopy and Diffuse Reflectance Infrared Fourier Transform Spectroscopy Insight into the Pathways of Manganese Oxalate Thermal Decomposition to MnO and MnCO₃

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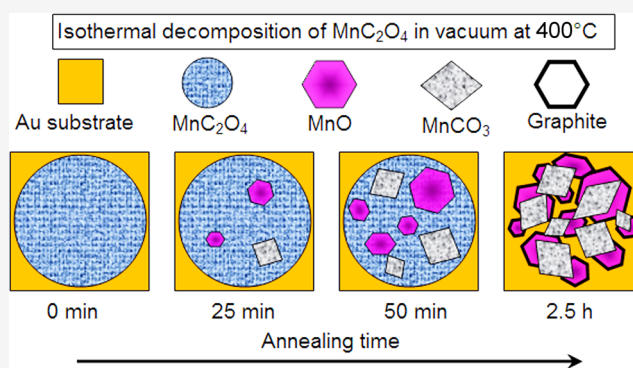
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ABSTRACT: X-ray photoelectron spectroscopy (XPS) and *in situ* diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) were employed to study the isothermal decomposition of MnC₂O₄ under ultrahigh vacuum and N₂ environmental conditions, respectively. High-resolution core-level XP spectra, X-ray-induced Auger spectra, and infrared spectra were obtained as a function of annealing time. In XPS studies, the time-dependent thermal decomposition characteristics were elucidated by analyzing surface composition, chemical shifts, satellites in the Mn 2p_{3/2} and Mn LMV bands, and Auger parameters for Mn and O. Functional groups developing during the ongoing reaction were identified by DRIFTS from characteristic vibrations. For the first time, the isothermal decomposition of manganese oxalate was shown to proceed *via* two pathways involving nucleation and accumulation of MnO and MnCO₃. The kinetics of the decomposition in vacuum could be described by the Prout–Tompkins or/and by the Avrami–Erofeev models. The results obtained by XPS, DRIFTS, and *ex situ* XRD allowed concluding that the final product of oxalate decomposition was composed of MnO and MnCO₃ or/and unidentate/polydentate carbonate structures populating the surface of the sample. A substantial formation of graphitic carbon was also observed and associated with interface chemical reactivities between the MnO particles and the supporting gold foil.



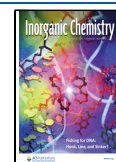
INTRODUCTION

Transition-metal oxalates MeC₂O₄·*n*H₂O with unfilled 3d electronic states are frequently compounds with a metal–organic framework-type structure and gain importance due to their potential technological applications in various fields such as the synthesis of nanostructures, catalysis, energy storage, and others. In particular, nanorods of manganese oxalate are being used as precursors to synthesize nanoparticles of various manganese oxides, including MnO, α-Mn₂O₃, and Mn₃O₄, through decomposition in different environments (vacuum, air, N₂).¹ Controlled heat treatment of MnC₂O₄·2H₂O in air allowed micro- and mesoporous manganese oxides with a narrow pore size distribution and high surface area (283 m²/g) to be obtained.² In fact, nanosized manganese oxides with a high surface area are known to exhibit good catalytic activity in various reactions. Accordingly, the thermal decomposition of manganese oxalate precursors was shown to be an effective way for synthesizing various support-free manganese oxide catalysts, both single phase^{3,4} and in combination with other metals and oxides,^{5,6} which demonstrate high activity in the oxidation of CO, volatile organic compounds, and CO hydrogenation.

The processes and factors governing the thermal decomposition of MnC₂O₄·2H₂O have been investigated for several decades. Typically, the thermal decomposition of Mn oxalate was performed in N₂^{7–9} or in air and N₂.^{10–15} Several studies were carried out in low vacuum^{14,16,17} and in He.¹⁸ A comparative analysis of the effect of different environments (N₂, CO₂, air, H₂) on the decomposition process was performed in ref 19. Usually, the decomposition of Mn oxalate was studied under nonisothermal conditions, albeit the isothermal decomposition at different temperatures was also explored and the type of kinetics defined.^{8,9,13,16,18} Generally, two main stages were observed in the thermal decomposition of MnC₂O₄·2H₂O that were associated with the dehydration and decomposition as such, with the characteristics of these stages being influenced by the environment and other factors. In all of these studies, traditional methods of thermal analysis,

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such as thermogravimetry (TG)/derivative thermogravimetry (DTG), differential thermal analysis (DTA), and differential scanning calorimetry (DSC), sometimes complemented by gas chromatography or mass spectrometric analysis of evolving gases, were employed. Infrared (IR) spectroscopy was used for *ex situ* characterization of the products obtained by calcination of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in air at 130, 250, 300, and 350 °C.¹⁰

As to the final product obtained by oxalate decomposition under different environments, there is considerable controversy. The thermal decomposition of Mn oxalate in air was reported to yield Mn_3O_4 ,^{10,11,13,15,19} Mn_2O_3 ,^{2,4,12,14,20} Mn_5O_8 , and a mixture of Mn_3O_4 + Mn_2O_3 ²¹ or Mn_3O_4 + Mn_5O_8 + MnO_2 .¹⁹ Such uncertainty in identification is related to the strong dependence of the structural state and composition of the final product on the heating rate, temperature, and time of annealing.^{2,4,13,19,21} *In situ* high-temperature X-ray diffraction (XRD) analysis revealed²⁰ that the decomposition of Mn oxalate in air started with the nucleation of amorphous MnO at lower temperatures, followed by its further oxidation at higher temperatures to Mn_2O_3 (or to Mn_3O_4 ¹³) and crystallization of the latter.

The solid residue obtained by decomposition of Mn oxalate in vacuum, N_2 , and CO_2 was assigned to MnO,^{7,8,11,12,14,17–19} mainly MnO¹⁵ and nonstoichiometric MnO with an admixture of Mn^{3+} ions¹⁶ or MnO + Mn_2O_3 .¹⁹ The presence of Mn^{3+} ions and Mn_2O_3 phase in the final product was explained by a small amount of impurity O_2 in the inert atmosphere and by the effect of the heating rate.¹⁹ Accordingly, there is a general agreement that the decomposition of anhydrous Mn oxalate in vacuum and in an inert environment proceeds according to $\text{MnC}_2\text{O}_4 \rightarrow \text{MnO} + \text{CO} + \text{CO}_2$,^{7,12–14} and the detected evolution of both CO and CO_2 ^{7,12,16,18} was claimed to substantiate this reaction scheme.

Note that the product of Mn oxalate thermal decomposition in vacuum and in an inert environment was commonly identified by TG mass loss (which always appreciably deviated from the theoretical value for MnO), *ex situ* XRD (which did not work in the case of amorphous oxide), *ex situ* IR spectroscopy, and even by visual observation of color. Obviously, these techniques cannot ensure an accurate determination of the actual composition of the residue, which is decisive for the clarification of the mechanism of oxalate decomposition. In the present work, X-ray photoelectron spectroscopy (XPS) is used for the first time to monitor physicochemical changes occurring during isothermal annealing of MnC_2O_4 under ultrahigh vacuum (UHV) conditions. As a surface-sensitive technique of chemical analysis, XPS provides direct information on the surface composition, chemical state of atoms, and electronic structure at different stages of oxalate decomposition which cannot be attained by bulk analysis techniques such as TG and XRD. Also, infrared spectroscopy is employed for the first time for *in situ* monitoring the chemical changes on the surface of Mn oxalate during its isothermal decomposition in an inert environment (N_2). The outcome of this work is believed to provide a profound understanding of the decomposition process and the final product formation, which will enable the controlled synthesis of dispersed manganese oxide powders with specific chemical surface characteristics favorable for catalytic applications.

EXPERIMENTAL SECTION

The synthesis procedure has been described in detail elsewhere.²² Briefly, to prepare manganese oxalate dihydrate, 0.5 mol/L acetic solutions of $\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ (100 mL) and oxalic acid $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (150 mL) were prepared in separate beakers. The solution of $\text{Mn}(\text{NO}_3)_2$ was added to the solution of $\text{H}_2\text{C}_2\text{O}_4$ under vigorous stirring. The precipitate was thoroughly washed with acetone, dried in air at 80 °C for 24 h, and ground. According to the XRD pattern taken with a Cu K α source using a Rigaku Miniflex-600 diffractometer, the as-synthesized oxalate powder is a $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ phase with a base-centered β -orthorhombic crystalline lattice (JCPDS #32-0646). In this structure, ligand ions $\text{C}_2\text{O}_4^{2-}$ are bound to metal ions with all four oxygen atoms in a bis-bidentate binding mode to form polymeric strands in which Mn ions are additionally coordinated with two H_2O molecules each (Figure 1).

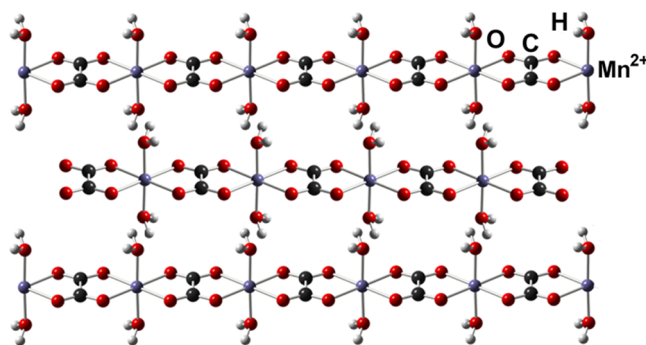


Figure 1. Representation of the molecular structure of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$. Note that in this polymeric metal–organic framework, carbon atoms in oxalate ligands are formally double-bonded ($\text{C}=\text{C}$).

The XPS study of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ thermal decomposition was performed in a combined XPS–time-of-flight secondary ion mass spectrometry (ToF SIMS) UHV instrument described elsewhere.²³ The oxalate powder was compacted in a “tray” of Au foil so as to form a dense layer suitable for XPS analysis, about 0.3 mm thick. The tray was fixed on a sample holder allowing indirect heating of the sample from the rear with a filament-type molybdenum heater up to 700 °C and temperature control with a Ni–NiCr thermocouple.²⁴ Prior to thermal treatments, as-prepared Mn oxalate was subjected to outgassing at room temperature in the preparation chamber at a base pressure of 5×10^{-10} mbar for about 80 h. For studying the thermal decomposition kinetics, the following procedure was employed. The outgassed oxalate sample was transferred to the reactor chamber with a base pressure of 9×10^{-10} mbar, heated with a rate of 5 °C/min to 180 °C, held at this temperature for 1 h for dehydration, and then heated with a rate of 50 °C/min to 400 °C at which temperature the sample was subjected to extended annealing for 25, 40, 55, 70, 110, and 150 min. The annealed sample was cooled down and transferred under ultrahigh vacuum to the analysis chamber with a base pressure of 2×10^{-10} mbar for taking XP spectra. As a reference for the initial state, XP spectra of as-prepared Mn oxalate at room temperature were taken as well.

High-resolution core-level XP spectra and Auger spectra were recorded with an energy step of 0.025 eV, using a hemispherical analyzer at a pass-energy of 50 eV and a nonmonochromatic Mg K α radiation at a power of 13 kV $\times$ 10 mA. After background subtraction (Shirley-type), the spectra were decomposed into their components with mixed Gaussian–Lorentzian lines (70% G + 30% L) by a nonlinear least squares curve-fitting procedure, using the CasaXPS software. The binding energy (BE) and full width at half-maximum (FWHM) of the components and their areas were determined from the fitting results. The surface composition of the samples was evaluated from the obtained peak areas and standard atomic sensitivity factors.²⁵ The surface-charging effects were taken into account using the O 1s reference line of oxalate ions set at 532.4 eV;²²

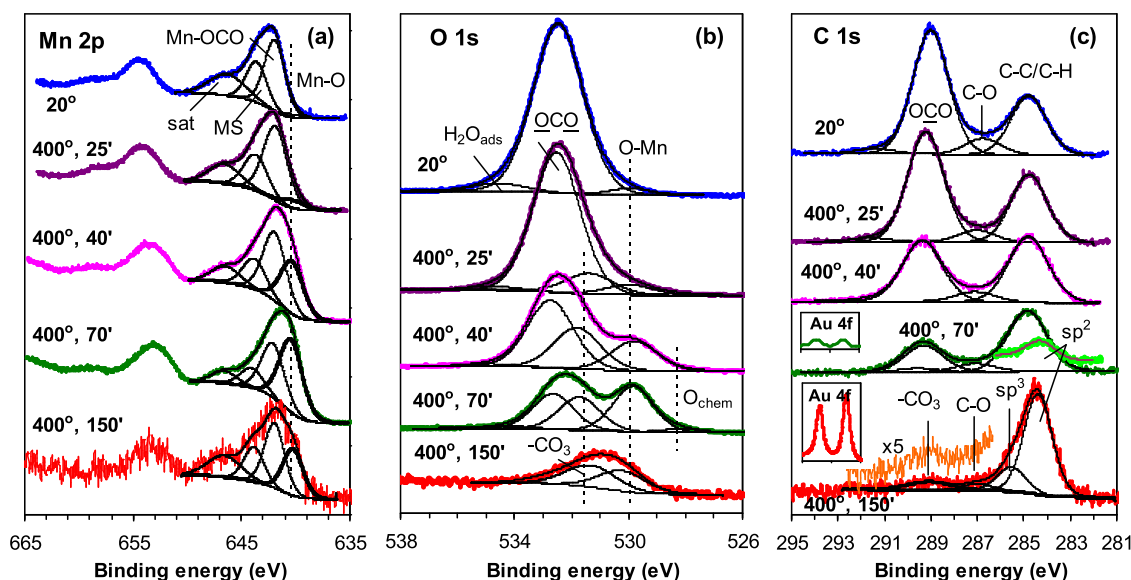


Figure 2. Core-level Mn 2p (a), O 1s (b), and C 1s (c) XP spectra of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in the as-prepared state (20 °C) and after annealing the oxalate at 400 °C for 25, 40, 70, and 150 min. The binding energy scale is charge-corrected. The deconvoluted components are indicated (multiplet splitting component and satellites are labeled “MS” and “sat”, respectively). Long-term annealing of the oxalate causes Au foil support to be exposed, as indicated in the insets of the respective C 1s spectra.

for cross-checking, the C 1s line of adventitious carbon at 284.8 eV was used. For the decomposed oxalate obtained by annealing at 400 °C for 150 min, all of the spectra were referenced to the O 1s line at 530.0 eV characteristic of MnO.

All core-level and Auger XP spectra and XRD pattern of the final product demonstrated low intensity because of a small amount of analyzed material caused by technical-methodological reasons. Due to intense gas evolution and shrinkage in the decomposition stage, the layer of the pressed oxalate powder cracked and crumbled. When transferring the decomposed sample from the reactor chamber through the distribution chamber to the analysis chamber for taking XP spectra, a certain amount of the crumbled material could get lost (swept away) both as a result of pumping and due to transfer-induced mechanical shaking.

Infrared spectra were collected in a diffuse reflectance mode using a Bruker Vertex 70 Fourier transform infrared (FTIR) spectrometer equipped with an MCT detector cooled by liquid nitrogen and an HT DRIFT Cell (Harrick Scientific) with ZnSe windows. The diffuse reflectance infrared Fourier transform (DRIFT) spectra were acquired by accumulating 1024 scans in the 700–4000 cm^{-1} spectral range with a resolution of 4 cm^{-1} . The sample was prepared by mixing the powder of as-synthesized $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (10% by mass, dried in air at 110 °C) with spectroscopic-grade KBr. To minimize the influence of gas-phase products (CO_2 , H_2O), the sample compartment was continuously purged with dry N_2 . The kinetics of thermal decomposition of Mn oxalate was studied *in situ* in the DRIFT cell as follows. First, the IR spectrum of a fresh sample was taken at room temperature. Then, under a nitrogen flow of 30 mL/min, the sample was heated at a rate of 10 °C/min up to 400 °C while recording the spectra every 50 °C. After attaining 400 °C, the sample was isothermally annealed until complete decomposition of the oxalate. At this temperature, IR spectra were taken every 3 min. A K-type thermocouple was placed in close proximity to the oxalate sample for accurate temperature control. For comparison, DRIFT spectra of the decomposed oxalate were also recorded after cooling to room temperature under flowing N_2 . A MnO sample prepared by *in situ* reduction of MnO_2 in 10% H_2 + He at 400 °C was used to provide a DRIFT reference.

RESULTS AND DISCUSSION

XP Spectral Analysis. Figure 2 demonstrates Mn 2p (a), O 1s (b), and C 1s (c) core-level photoelectron spectra of

$\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in the as-prepared state (20 °C) and after annealing at 400 °C for different times. Similar to our previous work,²² the broad Mn $2p_{3/2}$ band of the pristine oxalate was fitted with components representing the main Mn^{2+} state in the $-\text{OCO}-$ environment (at BE ~ 641.9 eV), multiplet splitting contribution, and shake-up satellite. Note that the intense satellite is indicative of a strong covalency of the $\text{Mn}^{2+}-\text{C}_2\text{O}_4^{2-}$ bonding in the oxalate.²² The O 1s band of the as-prepared oxalate includes a dominant component at BE = 532.4 eV associated with an $\text{O}-\text{C}-\text{O}$ environment and two minor components corresponding to $\text{O}-\text{Mn}$ bonds (~ 530.2 eV) and adsorbed H_2O (~ 534.5 eV). The C 1s band is also dominated by an oxalate-related component at ~ 289 eV ($\text{O}-\text{C}-\text{O}$ environment); besides, there are components at 284.8 and 286.7 eV attributed, respectively, to C–C/C–H bonds of adventitious carbon and C–O bonds arising from contamination of the oxalate by organics.

Figure 3 shows (a) Mn $\text{L}_{23}\text{M}_{23}\text{M}_{45}$ (LMV) and (b) O $\text{KL}_{23}\text{L}_{23}$ (KVV) X-ray-induced Auger electron spectra of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in the as-prepared state (20 °C) and after annealing the sample at 400 °C for different times. The Mn LVM spectrum of the fresh oxalate can be decomposed into four components, with two major components at kinetic energies of 585.8 and 581.6 eV being assigned to ^3P and ^1P spectral terms and two other components at lower kinetic energies representing a satellite structure as in Co oxalate.²⁶ The O KVV Auger line in the intact oxalate consists of two intense components separated by 4.5 eV, which are characteristic of the $\text{O}-\text{C}-\text{O}$ linkage.^{22,27}

Annealing of Mn oxalate at 400 °C causes notable transformations of the spectra. After holding the oxalate at this temperature for 25 min a new distinct component at ~ 640.4 eV emerges in the Mn $2p_{3/2}$ band; concurrently, in the O 1s band, the component assigned to $\text{O}-\text{Mn}$ bonds gains in intensity (Figure 2a,b). Besides, a new component arises in the O 1s band at ~ 531.6 eV (Figure 2b) which can be associated with oxygen atoms in the carbonate $-\text{CO}_3$ group.²⁸ With increasing annealing time up to 40 min fractions of these

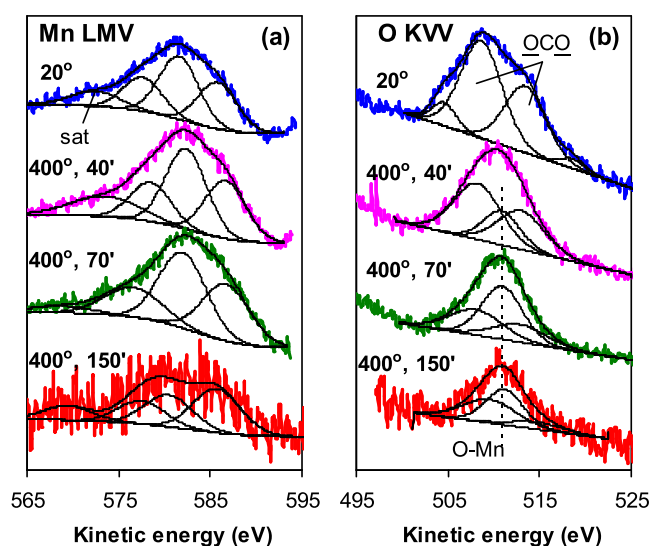


Figure 3. X-ray-induced Mn LMV (a) and O KVV (b) Auger electron spectra of $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in the as-prepared state (20 °C) and after annealing the oxalate at 400 °C for 40, 70 and 150 min. The kinetic energy scale is charge-corrected.

components in the Mn $2p_{3/2}$ and O $1s$ bands grow rapidly. Besides, a small additional component is detected in the O $1s$ band at ~ 528.2 eV after annealing for 40 min (Figure 2b) indicating the presence of chemisorbed or subsurface oxygen atoms. Also, a new distinct single-line component arises in the O KVV band at a kinetic energy (KE) of 511.1 eV (Figure 3b) implying the formation of an oxide phase.²⁷ Annealing for 70 min further promotes the fast intensity rise of these new components; at the same time, fractions of the oxalate-related components with Mn–OCO, $\text{O}=\text{C}=\text{O}$, and $\text{O}-\text{C}-\text{O}$ bonding in the respective Mn $2p_{3/2}$, O $1s$, C $1s$, and O KVV bands decrease monotonically (Figures 2a–c and 3b). Note also that in the C $1s$ band of the sample annealed for 70 min a new small component at ~ 289.6 eV with a fraction of $\sim 4\%$ can be resolved (Figure 2c). The Mn $2p_{3/2}$ BE = 640.4 eV, O $1s$ BE = 530 eV, and O KVV KE = 511.1 eV of newly evolving components are characteristic of those in manganese oxide MnO ,^{27,29} whereas the O $1s$ BE = 531.6 eV and C $1s$ BE = 289.6 eV along with their energy shift with respect to the O $1s$ OCO and C $1s$ OCO oxalate peaks indicate the formation of carbonate, in accordance with assignments for CoCO_3 and CoC_2O_4 in previous studies.²⁸ Accordingly, the concurrent intensity growth of the additional peaks along with the diminution of the oxalate-related components imply the nucleation and accumulation of the new phases MnO and MnCO_3 as a result of the isothermal decomposition of MnC_2O_4 .

During annealing, a fraction of the C–C/C–H component in the C $1s$ band monotonically increases (Figure 2c) from ~ 29 to 54% indicating out-diffusion of hydrocarbon contaminations and their accumulation on the surface before desorbing into the vacuum.²⁸ Note that after annealing for 70 min, another distinct C $1s$ peak appears at ~ 284.3 eV (Figure 2c), which points to the formation of graphitic carbon. As in the case of isothermal decomposition of cobalt oxalate,²⁶ the deposition of graphitic carbon on the decomposed product could be expected to accrue from the dissociation of evolving CO accompanied by the Boudouard reaction ($2\text{CO} \rightarrow \text{C}_{\text{ads}} + \text{CO}_2\uparrow$). However, in contrast to metallic Co which can catalyze

CO disproportionation,²⁶ the Boudouard reaction does not occur over MnO ^{30,31} which was formed in the course of Mn oxalate decomposition. In this case, the deposition of graphitic carbon is deemed to occur mainly *via* dissociation of CO at the interface between the MnO particles and the gold foil serving as a support for the oxalate powder. This is substantiated by the following facts: (i) in contrast to the C–C/C–H component representing hydrocarbon contamination of the oxalate, the graphitic C $1s$ peak is not shifted to higher BEs by surface charging, which indicates the deposition of graphite to occur on a metal, *i.e.*, Au support in our case; (ii) due to partial Mn oxalate decomposition, the sample somewhat shrinks, thus uncovering small areas of the gold foil underneath so that low-intensity Au 4f electron emission is detected (see the corresponding inset in Figure 2c). It has been shown^{32,33} that nanoparticles of oxides (*e.g.*, FeO_x , Fe_2O_3 , MgO , TiO_2 , CeO_2) deposited on a gold surface can effectively dissociate CO at the oxide/metal interface producing graphitic carbon and work as excellent ‘inverse catalysts’ for room-temperature CO oxidation. More than that, Li *et al.*³⁴ have recently demonstrated that the edge sites of MnO nanosized islands deposited on Au(111) provide active sites for CO dissociation which occurs at the MnO/Au interface above 200 °C producing surface carbon. Accordingly, during the thermal decomposition of Mn oxalate, the interface between the nascent MnO nanoparticles and the gold support is believed to also enable graphite deposition. Note that the intensity ratio of the spin–orbit components in the Au 4f spectrum recorded after annealing of the oxalate for 70 min (Figure 2c), $\text{Au } 4f_{7/2}/4f_{5/2} = 0.94$, is appreciably lower than the statistical ratio of the corresponding core-hole state degeneracies (4:3) and lower than the peak ratio $\text{Au } 4f_{7/2}/4f_{5/2} = 1.27$ measured for the pure Au substrate, indicating strong interaction of nascent MnO nanoparticles with the Au support, which causes alterations in the local Au electronic band structure.³⁵

Annealing for 2.5 h brings about complete decomposition of the oxalate. As a consequence, the characteristic OCO components are not detected in the O $1s$ and C $1s$ spectra anymore (Figure 2b,c). The Mn $2p_{3/2}$ band of the decomposed oxalate is rather wide (3.9 eV versus 3.0 eV for the fresh oxalate) and characterized by a high-BE hump at ~ 646.7 eV which is obviously related to a shake-up satellite (Figure 2a). The O $1s$ line of the final product is also rather broad; its FWHM is 2.7 eV compared to 2.0 eV for the fresh oxalate. It can be decomposed into two major components with fractions of 46 and 54% being located at 530.2 and 531.3 eV, respectively (Figure 2b). The C $1s$ spectrum of the final product shows up as an intense asymmetric band with a major component peaking at ~ 284.3 eV that indicates a significant deposition of graphitic carbon. Its content amounts to ~ 43 atom % which is about 5 times higher than that deposited after annealing for 70 min. This is associated with an outgrowth of the MnO/Au interface boundaries and is evidenced by an 8-fold increase in the intensity of the Au 4f peak (see inset in Figure 2c) due to uncovering larger areas of the gold substrate because of the sample shrinkage. In addition, a rather intense C $1s$ component (with a fraction of about 15%) located at BE = 285.3 eV indicates the presence of a deposited sp^3 hybridized carbon. Other minor C $1s$ components detected at higher BE are related to C–O (~ 287 eV) and $-\text{CO}_3$ bonding (~ 289.1 eV).

Time Dependences. To get a more detailed insight into changes occurring in the physicochemical state and electronic

structure of Mn oxalate in the course of its isothermal decomposition, the variation of spectral parameters with time of annealing was analyzed. Figure 4a shows the evolution of the

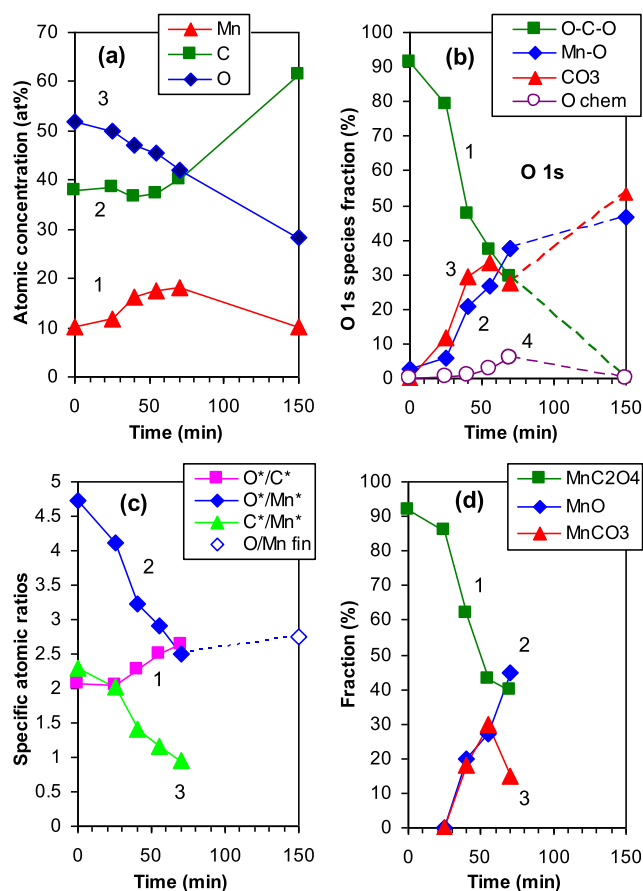


Figure 4. (a) Atomic surface concentration of the main constituents Mn (1), C (2), and O (3) of manganese oxalate; (b) fractions of oxygen species related to $\text{O}-\text{C}-\text{O}$ (1), $\text{O}-\text{Mn}$ (2), and $-\text{CO}_3/\text{O}=\text{C}$ (3) bonding and chemisorbed O (4) in the O 1s band; (c) specific atomic ratios O^*/C^* (1), O^*/Mn^* (2), and C^*/Mn^* (3) for the components related to OCO environment in the respective O 1s, C 1s, and Mn 2p spectra; (d) evaluated fractions of residual MnC_2O_4 (1), MnO (2), and MnCO_3 (3) after emission of x CO and y CO_2 molecules (see text for details) as a function of time during isothermal annealing of MnC_2O_4 at 400 °C.

general surface chemical composition of manganese oxalate represented by the total concentration of manganese, carbon, and oxygen as evaluated from the respective spectra presented in Figure 2 during annealing at 400 °C. Figure 4b shows the fractions of various oxygen species derived in a four-peak fit of the O 1s band as a function of the annealing time. It can be seen that the fraction of oxalate-related species, $\text{O}-\text{C}-\text{O}$, rapidly decreases, whereas the fraction of oxygen atoms with $\text{O}-\text{Mn}$ bonding related to manganese oxide monotonically increases. The fraction of oxygen species ascribed to a carbonate group rises nonmonotonically, demonstrating some diminution after annealing for 70 min. Note also the increasing fraction (up to ~5.8% after 70 min) of oxygen atoms chemisorbed on the surface or in the subsurface layer of nascent MnO.

Using areas of the dominant components Mn^* , O^* , and C^* related to the OCO environment in the respective Mn 2p (including multiplet and satellite features), O 1s, and C 1s XP spectra and standard sensitivity factors, the specific atomic

ratios O^*/C^* , O^*/Mn^* , and C^*/Mn^* were evaluated. For as-prepared Mn oxalate, the ratios $\text{O}^*/\text{C}^* = 2.06$, $\text{O}^*/\text{Mn}^* = 4.67$, and $\text{C}^*/\text{Mn}^* = 2.29$ were obtained which correspond to a stoichiometric oxalate containing also contaminating oxalic acid in the amount of ~0.15 $\text{H}_2\text{C}_2\text{O}_4$ molecules not involved in ligand bonding.²² With increasing time of annealing up to 70 min, the O^*/Mn^* and C^*/Mn^* ratios rapidly decrease, whereas the O^*/C^* ratio increases to 2.6 (Figure 4c). Note that this unusual behavior of the O^*/C^* ratio was not observed in the course of thermal decomposition of Ni²³ and Co²⁶ oxalates. To comprehend this trend, we assume that the decomposition of Mn oxalate could proceed via two pathways: $\text{MnC}_2\text{O}_4 \rightarrow \text{MnO} + \text{CO} + \text{CO}_2$ and $\text{MnC}_2\text{O}_4 \rightarrow \text{MnCO}_3 + \text{CO}$ since the process of decomposition is known^{7,16,18} to be accompanied by the evolution of CO and CO_2 . Consequently, after emission of x CO and y CO_2 molecules the surface composition of the products formed at a given moment can be represented as $\text{MnC}_2\text{O}_4 - x\text{CO} - y\text{CO}_2 = (1-x)\text{MnC}_2\text{O}_4 + y\text{MnO} + (x-y)\text{MnCO}_3$. We determined x and y by adjusting the calculated O^*/C^* , O^*/Mn^* and C^*/Mn^* ratios to the measured ones (Figure 4c, curve 1). The derived fractions of residual MnC_2O_4 , MnO, and MnCO_3 are presented in Figure 4d. One can see that variations of these fractions as a function of annealing time are qualitatively very similar to those of the deconvoluted fractions of respective oxygen species $\text{O}-\text{C}-\text{O}$, $\text{O}-\text{Mn}$, and $-\text{CO}_3$ (Figure 4b) and thus support their assignment. Note even a consistent decrease in the content of MnCO_3 after annealing of the oxalate for 70 min. Such a decrease can be explained by a competing process of carbonate decomposition at this temperature.^{36,37}

As can be seen in Figure 4b, the fraction of $\text{O}-\text{Mn}$ species in the O 1s band (curve 2) associated with manganese oxide demonstrates a sigmoidal-type dependence on the decomposition time t . Accordingly, we analyzed this dependence on the basis of different kinetic models. It was found that the kinetics of Mn oxalate isothermal decomposition in vacuum represented by fractions α of the $\text{O}-\text{Mn}$ species in the O 1s band as well as by fractions α of the $\text{Mn}-\text{O}$ species in the Mn 2p_{3/2} band (see Figure 2a,b) could be most satisfactorily described—as in the case of isothermal decomposition of cobalt oxalate in vacuum²⁶—by the Prout–Tompkins rate

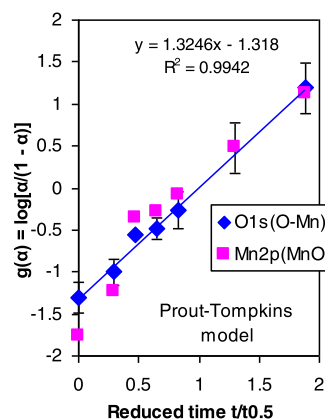


Figure 5. Fraction α of formed manganese oxide represented by a fraction of the $\text{O}-\text{Mn}$ species in the O 1s band and by a fraction of the $\text{Mn}-\text{O}$ species in the Mn 2p_{3/2} band versus reduced time of Mn oxalate isothermal decomposition plotted for Prout–Tompkins kinetic model.

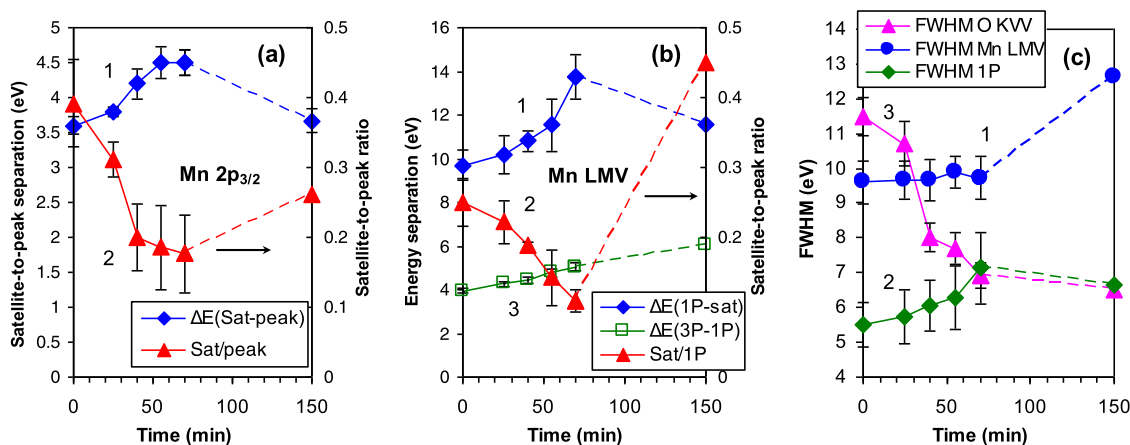


Figure 6. (a) Energy separation ΔE between the satellite and the peak (1) and satellite-to-peak intensity ratio (2) in the Mn $2p_{3/2}$ band (see Figure 2a); (b) energy separation ΔE between the most intense term $1P$ and the satellite (1), satellite-to-peak intensity ratio (2), and energy separation ΔE between the terms $3P$ and $1P$ (3) in the Mn LMV Auger band (see Figure 3a); and (c) FWHM of the O KVV (1) and Mn LMV (2) Auger bands and FWHM of the $1P$ component in the Mn LMV band (3) as a function of time during isothermal annealing of MnC_2O_4 at 400 °C.

equation (Figure 5) for an autocatalytic reaction of topochemical decomposition

$$g(\alpha) = \lg[\alpha/(1 - \alpha)] = k_B \tau$$

where $\tau = t/t_{0.5}$ is the reduced time, with $t_{0.5}$ being the time of half-decomposition ($\alpha = 0.5$), and k_B is the reaction rate constant. Alternatively, our data for the manganese oxalate decomposition in vacuum can be just as well described by the Avrami–Erofeev rate equation $g(\alpha) = [-\ln(1 - \alpha)]^{1/n} = k\tau$ with $n = 4$, which corresponds to a three-dimensional growth of spherical (or semispherical) nuclei, albeit with a slightly worse correlation ($R^2 = 0.9914$) than with the Prout–Tompkins equation ($R^2 = 0.9942$).

In previous works, the kinetics of isothermal decomposition of anhydrous manganese oxalate in N_2 studied by TG in the temperature range 385–425 °C was described by a “contracting sphere equation”, $1 - (1 - \alpha)^{1/3} = kt$,¹³ and by first-order functions, $1 + \log(1 - \alpha) = kt$ and $-\ln(1 - \alpha) = kt$.^{8,9} The contracting sphere equation was also used to describe the kinetics of manganese oxalate isothermal decomposition in vacuum by measuring yields of CO and CO_2 at 363 °C.¹⁶ In a He environment, the TG-derived kinetics of isothermal decomposition at 338–350 °C was reported¹⁸ to be of the Avrami–Erofeev-type $[-\ln(1 - \alpha)]^{1/n} = kt$ with $n = 1.9$.

High-temperature annealing of Mn oxalate causes a variation of the Mn $2p$ spectral parameters. The peak-to-peak spin–orbit splitting of the Mn $2p$ line demonstrates a slight decrease from 11.95 eV for fresh oxalate to 11.73 eV for the final product. More appreciable changes occur in the Mn $2p_{3/2}$ peak-to-satellite energy separation, which monotonically increases from 3.6 eV for fresh oxalate to 4.5 eV for the sample annealed for 70 min, before dropping to 3.7 eV in the decomposed product (Figure 6a). Contrariwise, the satellite-to-Mn $2p_{3/2}$ peak intensity ratio monotonically decreases with increasing time of annealing, and after completing the decomposition, the ratio increases again (Figure 6a). Our data obtained for the series of transition-metal Me^{2+} oxalates and MeO oxides as well as the analysis of the reported data for the series of transition-metal Me^{2+} compounds with different oxygen-containing anions and Me–O bonds show that the satellite separation steadily increases and the relative satellite intensity generally decreases

with increasing ionicity of the Me–O bond.²² Therefore, the concordant increase in the satellite-to-Mn $2p_{3/2}$ peak energy separation along with a decrease in the satellite-to-Mn $2p_{3/2}$ peak intensity ratio may indicate an increase in ionicity of the –O–Mn–O– bonds with increasing time of annealing. This is consistent with the observed growing abundance of MnO particles which have more ionic Mn–O bonds compared to Mn–O₄C₂ bonds in the oxalate. A similar increase in the ionicity of Co–O bonds was observed in decomposing Co oxalate upon nucleation of CoO.²⁶ The decelerating augmentation at longer annealing times followed by a decrease in the satellite-to-peak separation and an increase in the satellite-to-peak ratio in the decomposed Mn oxalate may reflect competing accumulation of more covalent $MnCO_3$.

Appreciable changes in spectral parameters indicating chemical and phase transformations in the oxalate are also observed in the Mn LMV and O KVV Auger bands. As can be seen in Figure 6b, the energy separation between the satellite and the most intense $1P$ term and the satellite-to-peak intensity ratio in the Mn LMV band vary with annealing time in a qualitatively similar way as those in the Mn $2p_{3/2}$ band (cf. Figure 6a). Besides, the energy distance between the terms $3P$ and $1P$ in the Mn $2p_{3/2}$ band steadily increases from 3.95 eV in the as-prepared oxalate to 6.01 eV in the decomposed product (Figure 6b, curve 3). In the course of annealing, no appreciable broadening of the Mn LMV band occurs (Figure 6c, curve 1), though the FWHM of the components attributed to $1P$ and $3P$ terms gradually increases from 5.5 to 7.1 eV (Figure 6c, curve 2). In the decomposed product, the FWHM of the Mn LMV band steeply increases to 12.6 eV. On the contrary, the FWHM of the O KVV band significantly decreases with annealing time from 11.5 eV for the as-prepared sample to 6.5 eV for the decomposed oxalate (Figure 6c, curve 3) due to a decreasing contribution of the broad oxalate-related doublet and increasing contribution of the narrow MnO-related peak (cf. Figure 3b).

Using the peak binding energies of the O $1s$ and Mn $2p_{3/2}$ photoelectron lines and the kinetic energies of the most intense components in the respective O KVV and Mn LMV bands ($1D$ and $1P$ spectral terms, respectively), the modified Auger parameters $\alpha'(O) = BE(O\ 1s) + KE(O\ KVV)$ and $\alpha'(Mn) = BE(Mn\ 2p_{3/2}) + KE(Mn\ LMV)$ were evaluated. For

the as-prepared $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$, $\alpha'(\text{O}) = 1041.0$ eV and $\alpha'(\text{Mn}) = 1223.8$ eV. Figure 7 shows that with increasing time

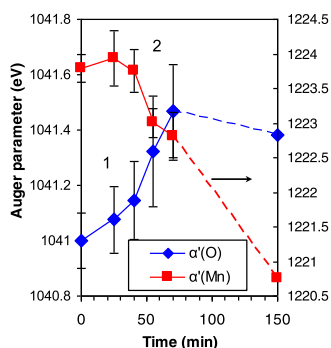


Figure 7. Auger parameters for oxygen $\alpha'(\text{O})$ (curve 1) and manganese $\alpha'(\text{Mn})$ (curve 2) versus time of isothermal annealing of MnC_2O_4 at 400 °C.

of annealing $\alpha'(\text{O})$ gradually increases before dropping to the value characteristic of the final product (curve 1); on the contrary, the $\alpha'(\text{Mn})$ parameter nonmonotonically decreases (curve 2). This demonstrates the sensitivity of the Auger parameters to alterations in the local chemical and structural states of O and Mn atoms and indicates the varying environment-dependent polarizability of these atoms in the oxalate upon its decomposition.

Final Product. To determine the chemical state of manganese in the final product, the Mn $2p_{3/2}$ band was tentatively fitted with 3 components located at 640.3, 641.9, and 643.8 eV and a satellite (Figure 2a). Such three-peak fitting of the main Mn $2p_{3/2}$ band and assignment of these components to Mn^{2+} , Mn^{3+} , and Mn^{4+} species, respectively, are commonly used for deriving oxidation states of Mn in catalysts containing manganese oxides after calcination in air, as, for instance, in refs 4, 38. In our case, the average oxidation state of Mn evaluated from the above components amounts to ~ 2.7 , implying the surface composition of the decomposed product to be close to 30% MnO + 70% Mn_2O_3 . According to the

literature,^{1,14,16,17} the final product of the Mn oxalate thermal decomposition in vacuum is MnO , although the appearance of Mn^{3+} as an intermediate in the reaction $\text{MnO} + 1/2 \text{CO}_2 \leftrightarrow 1/2 \text{Mn}_2\text{O}_3 + 1/2 \text{CO}$ has been admitted.¹⁶ However, such a large fraction of Mn_2O_3 ($\sim 70\%$) can hardly be expected to form by reaction with nascent CO_2 under UHV conditions. Even more so, the occurrence of a highly intense satellite in the Mn $2p_{3/2}$ band (and of an intense concordant satellite in the Mn LMV spectrum of the final product—Figure 3a) provides clear evidence that Mn^{3+} cannot be the dominant component in the final product. Note that no distinct shake-up satellite is observed in the Mn $2p_{3/2}$ band of Mn_2O_3 .^{29,39} On the other hand, an intense shake-up satellite can be seen in the Mn $2p_{3/2}$ band of MnCO_3 .^{40,41}

Following the approach suggested by Biesinger et al.³⁹ (which may be regarded debatable in view of the large number of contributing components), we fitted the Mn $2p_{3/2}$ band of the decomposed oxalate with (i) two sets of components corresponding to MnO (5 lines plus satellite) and Mn_2O_3 (5 lines) (Figure 8a) and (ii) three sets of components corresponding to MnO (5 lines and satellite), Mn_2O_3 (5 lines) and MnCO_3 (3 lines and two satellites) (Figure 8b). In view of the similarity between the Mn $2p_{3/2}$ and Fe $2p_{3/2}$ bands in respective carbonates, for fitting MnCO_3 , we made use of the set of lines available for FeCO_3 , having preserved all of the fitting parameters, i.e., the number of components, their FWHM, energy separation and intensity.³⁹ The sets of lines corresponding to MnO and Mn_2O_3 fail to properly fit the Mn $2p_{3/2}$ band of the final product (Figure 8a). On the contrary, an adequate fitting is achieved using the sets of lines for MnO and MnCO_3 (Figure 8b). Note that the intensity of the Mn_2O_3 -related lines included in the fitting (ii) always went to zero for all fitting trials. Thus, according to the fitting results presented in Figure 8b, the surface layer of the final product is composed of MnO and MnCO_3 , with no Mn^{3+} species being present. The Mn 3p spectrum peaking at 47.6 eV (Figure 8c) corroborates the average oxidation state of Mn in the final product to be 2+. ⁴² Now, the components of the simple fit in Figure 2a can alternatively be interpreted as being related to

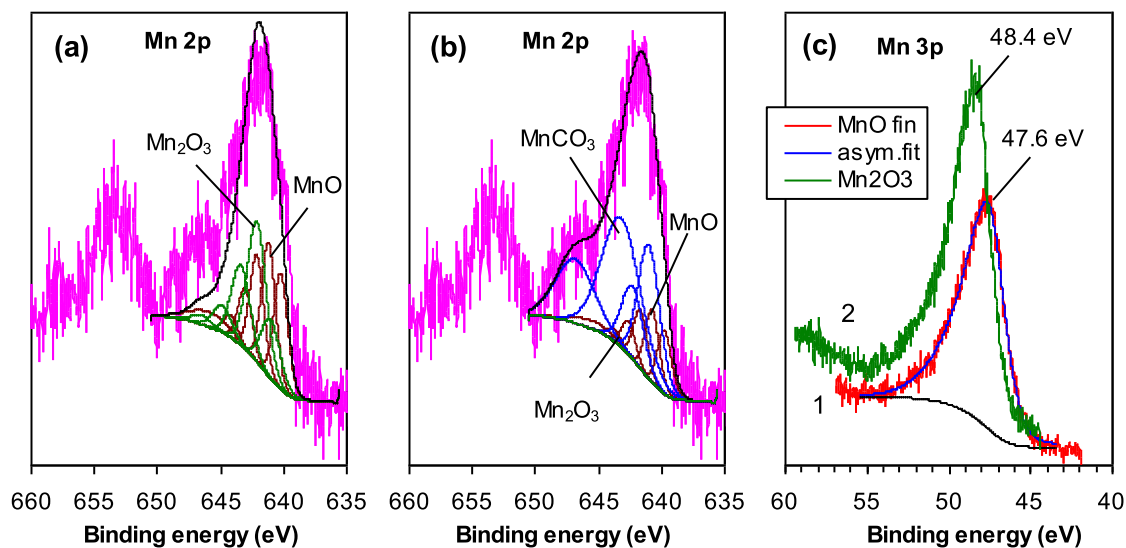


Figure 8. Multicomponent fitting of the Mn $2p_{3/2}$ band for the decomposed oxalate (400 °C, 150 min) using a combination of spectral fitting parameters³⁹ corresponding to (a) MnO and Mn_2O_3 and to (b) MnO , Mn_2O_3 and MnCO_3 ; (c) Mn 3p spectra of the final product (1) and Mn_2O_3 (2).

Mn²⁺ ions in MnO (640.3 eV) and Mn²⁺ ions in MnCO₃ (641.9 eV) along with a multiplet splitting contribution (643.8 eV) and MnCO₃-related shake-up satellite (646.7 eV).

The ionicity of the Mn–O bonds characterized by the satellite-to-peak energy separation and satellite-to-peak intensity ratio in the Mn 2p_{3/2} band is observed to steeply decrease in the final product (Figure 6a). This implies an increased contribution of a more covalent compound and corroborates the dominant abundance of surface MnCO₃. Accordingly, the two major components in the O 1s spectrum (Figure 2b) can be assigned to different states of oxygen ions in MnO (530 eV) and, predominantly, in MnCO₃ (531.2 eV) with fractions of 46.4 and 53.6%, respectively (Figure 4b). The total atomic ratio O/Mn = 2.75 evaluated from areas of the O 1s and Mn 2p_{3/2} bands for the final product (Figure 4c) turns somewhat larger than the ratio O/Mn = 2.42 corresponding to the derived surface composition because of adsorbed oxygenated carbon species with O–C, O=C bonding contributing to the O 1s spectrum. The occurrence in the C 1s spectrum of a component at ~289.1 eV associated with carbonate species –CO₃ is also consistent with the formation of surface MnCO₃. Note that the energy separation between the components related to OCO bonding in the O 1s and C 1s bands, ΔBE(OCO), steadily decreases with time of annealing from the characteristic value of 243.5 eV for the as-prepared oxalate to 242.1 eV for the final product, indicating a gradual transformation of the oxalate to carbonate.²² Also, the atomic ratio O/C for CO₃-related components in the O 1s and C 1s spectra of the final product was evaluated to be about 2.8, which is close to that in carbonate.

Ex situ powder XRD analysis of the final product obtained by decomposition of MnC₂O₄ in UHV at 400 °C for 150 min provides additional evidence for crystalline MnCO₃ formation. According to the XRD pattern (Figure 9), the final product is

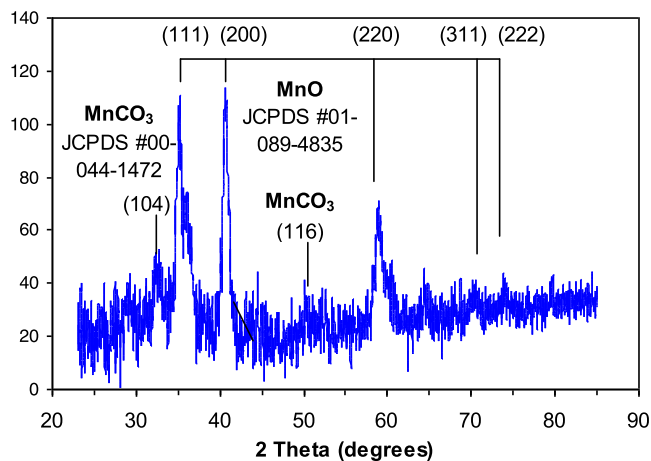


Figure 9. XRD pattern (Cu Kα) of a final product of the MnC₂O₄ decomposition in UHV at 400 °C for 150 min.

composed of predominantly MnO having a rock salt structure (JCPDS #01-089-4835) and a small amount of carbonate MnCO₃ with a calcite-type rhombohedral structure (JCPDS #00-044-1472).

In previous studies, the decomposition of Mn oxalate in an inert environment (N₂, He) and in vacuum was considered to proceed *via* a pathway (1) MnC₂O₄ → MnO + CO + CO₂, and the formation of MnO as a final product was derived from TG and XRD measurements. Using XPS, we show directly for

the first time that the decomposition of MnC₂O₄ in vacuum occurs through nucleation and accumulation of two phases, MnO and MnCO₃, and provide clear evidence for the opening of an additional pathway (2) MnC₂O₄ → MnCO₃ + CO. Bakcsy and Hegedüs suggested¹⁴ that pathway (2) could be highly feasible since the change of enthalpy in the decomposition reaction (2) occurring in an inert atmosphere and vacuum with the formation of MnCO₃, ΔH₂₉₈ = +4 kcal/mol, is significantly less than the change of enthalpy in the reaction (1) with formation of MnO, ΔH₂₉₈ = +29 kcal/mol, whereby the relative amount of the carbonate is determined by the competition of formation and thermal decomposition processes. This is exactly what we observe experimentally. Note that while decomposing Mn oxalate in air, MnCO₃ was observed to form as an intermediate followed by its final transformation into Mn₃O₄.¹⁰

The XPS-derived surface composition of the final product obtained by annealing the oxalate at 400 °C for 2.5 h is characterized by a large amount of MnCO₃ which significantly exceeds that of MnO. The predominant abundance of MnCO₃ on the surface could ensue from the local long-term interaction of MnO with CO₂ as a product molecule being concurrently released in the decomposition process (1) to form the carbonate MnO + CO₂ ↔ MnCO₃.³⁶ Note that the formation of a carbonate on the MnO/Au(111) surface was observed after exposure to 10 mbar CO₂ even at room temperature.³⁴ Besides, MnCO₃ was reported to form, for instance, in FeO–MnO,⁴³ Co–MnO,⁴⁴ and carbon nanotubes–MnO_x nanocomposite⁴⁵ catalysts under reaction conditions of the long-term Fischer–Tropsch synthesis at 260–300 °C in a mixture of CO + H₂. With all of these catalysts, a considerable water–gas shift activity was observed, possibly involving a formate/carboxylate-derived complex with kinetic branching leading to either chain lengthening *via* hydrogenation or CO₂ spin-off. No doubt, MnO would be easily converted to MnCO₃ under these circumstances. From the viewpoint of catalytic performance, the reaction-induced formation of MnCO₃ in MnO-containing catalysts was considered to be both favorable⁴³ and detrimental.⁴⁵ A clear-cut decision of its catalytic action, inhibition, or spectatorship cannot be made at present and is certainly also dependent on the catalytic process under consideration. Accordingly, the results of our study revealing the formation of manganese carbonate during the thermal activation of a Mn oxalate precursor, either pure or as a co-precipitate, may trigger further investigations to elucidate the role such carbonates may play in catalysis.

Infrared Spectroscopic Analysis. To provide additional evidence supporting our XPS findings, we performed an *in situ* IR spectroscopic study of the nonisothermal and isothermal decomposition of Mn oxalate in N₂. The DRIFT spectrum of as-prepared MnC₂O₄·2H₂O (Figure 10, 20°) exhibits all of the characteristic features of an oxalate hydrate:⁴⁶ a broad band in the range 3300–3700 cm^{−1} with an absorption maximum at 3618 cm^{−1} corresponds to the –OH stretching vibrations of coordinated water; an intense band centered at ~1622 cm^{−1} (with a side feature at 1707 cm^{−1}) results from asymmetric stretching of the O–C–O groups in the oxalate structure and is broadened by a contribution of the HOH bending mode; two intense lines at 1361 and 1323 cm^{−1} associated with the symmetric stretching vibrations of the O–C–O groups indicate the presence of bidentate oxalate groups bound with all four oxygen atoms to surface metal atoms; an intense narrow band at 791 cm^{−1} corresponds to the asymmetric

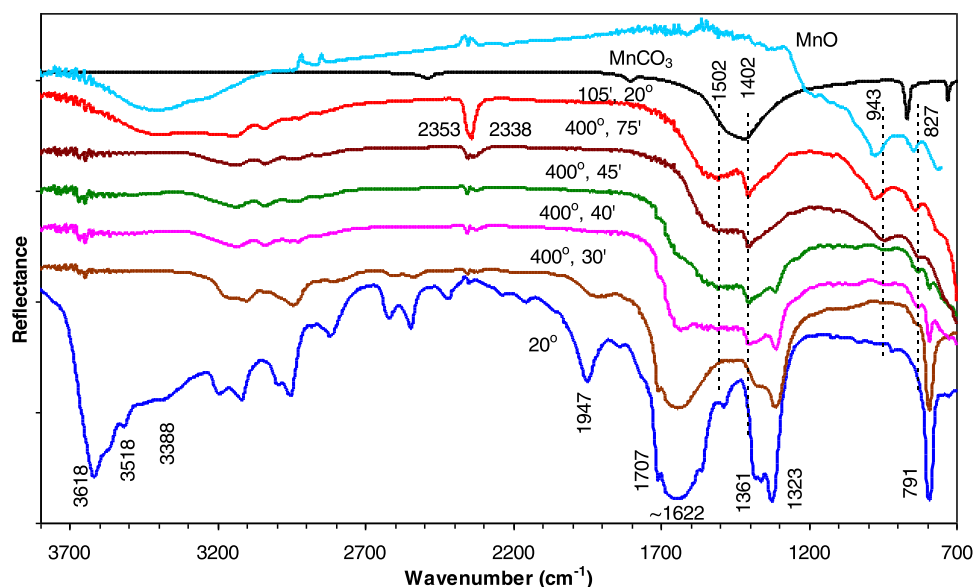


Figure 10. DRIFT spectra taken for $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in the as-prepared state (20 °C) and during isothermal annealing in N_2 flow at 400 °C for 30, 40, 45, and 75 min. The spectrum recorded after annealing the oxalate at 400 °C for 105 min and cooling down to room temperature under N_2 flow is also shown. The reference spectra of bulk MnCO_3 and MnO are presented in the top part of the figure for comparison. The spectra are shifted along the y-axis for the sake of clarity.

bending mode $\delta_{\text{as}}(\text{OCO})$. The absorption band at 1947 cm^{-1} may arise from a combination of vibrations (combination band).

Heating the oxalate to 150 °C causes an appreciable diminution of the H_2O -related band at 3618 cm^{-1} , which totally vanishes after annealing at 150 °C for 1 h or after nonisothermal heating to 250 °C indicating completion of the dehydration stage. The IR spectra of anhydrous Mn oxalate taken ~30 min after attaining the temperature of isothermal annealing (400 °C) display the same characteristic bands as in the dihydrate that correspond to asymmetric and symmetric stretching and bending vibrations of the OCO groups; the only difference being a notable narrowing of the $\nu_{\text{as}}(\text{OCO})$ band and a shift to lower frequencies (1311 cm^{-1}) of $\nu_{\text{s}}(\text{OCO})$ (Figure 10, 30 min).

After isothermal annealing the oxalate for ca. 35–40 min, new absorption bands arise at 827, 943, and 1402 cm^{-1} (Figure 10, 40 min). Further annealing for ~45 min produces an additional low-intense broad band at ~1500 cm^{-1} (Figure 10). Annealing for ≥ 60 min results in vanishing oxalate-related bands (Figure 10, 75 min). The IR spectra of the final product produced by annealing for 60–105 min are characterized by a set of pronounced absorption bands at 827, 943, 1402, and ~1502 cm^{-1} . The IR spectrum of the sample annealed at 400 °C for 105 min and cooled down to room temperature demonstrates a significant enhancement of these bands, with the former two being appreciably shifted to higher frequencies of 839 and 976 cm^{-1} , respectively (Figure 10).

Note that the oxalate decomposition is accompanied by the appearance and monotonic growth of an absorption band composed of two components at 2338 and 2353 cm^{-1} related to the gas-phase CO_2 . This doublet, however, becomes rather intense after cooling the sample to room temperature (Figure 10). Similar two-peak bands at 2340–2345 and 2353–2358 cm^{-1} were observed by Lercher et al.⁴⁷ upon adsorption of CO_2 on Al_2O_3 , $\text{Al}_2\text{O}_3/\text{MgO}$, and MgO oxides and were attributed to asymmetric stretching vibrations of a linearly adsorbed CO_2 molecule interacting *via* one oxygen atom with

coordinatively unsaturated surface metal cations of somewhat different polarizing strength. Also, an intense single band at 2340 cm^{-1} and a doublet at 2340 and 2356 cm^{-1} were observed, respectively, in thermally treated carbonate-containing apatites and powdered enamel. These bands were associated with the CO_2 arising from the decomposition of the compound and its occlusion in the solid.⁴⁸ In our experiments, CO_2 evolving during the course of the manganese oxalate decomposition could also be trapped either in pores and cavities formed by inhomogeneous oxalate decomposition (polymeric strands do not break all at once) or in the free space of the texture spanned by adjacent crystallites of the final product. At high temperatures, CO_2 may then diffuse outward and desorb, followed by readsorption upon cooling to room temperature so as to produce the observed intense two-peak band.

The bands at ~1502 and 1402 cm^{-1} unequivocally indicate the formation of a carbonate complex. Indeed, similar bands at 1500–1550 and 1400–1420 cm^{-1} were observed to arise during adsorption of CO_2 on MgO and CaO ,⁴⁹ MgO/CaO mixed oxides⁵⁰ and Al_2O_3 , MgO and $\text{Al}_2\text{O}_3/\text{MgO}$ mixed oxides⁴⁷ and were identified as resulting from the adsorption-induced splitting of a carbonate ion asymmetric stretching vibration ν_3 into, respectively, asymmetric and symmetric stretching vibrations of the O_1CO_1 bonds. The frequency splitting $\Delta\nu_3$ of about 100 cm^{-1} between the two vibrations was suggested to be indicative^{47,49–51} of unidentate carbonate bonding. However, low $\Delta\nu_3$ values do not unambiguously identify unidentate structures, especially on oxides with low polarizing power of the cation (such as Mn^{2+} in MnO , as in our case), and can be observed for carbonate ions in metal complexes with all three carbonate oxygens bound to metal ions (bridged polydentate structures).⁵¹

A more accurate identification of carbonate species with small $\Delta\nu_3$ splitting can possibly be achieved by considering their thermal stability. Though a bidentate structure is typically more strongly bound to the surface than a unidentate one, the resistance of the latter to thermal decomposition is still rather

high.⁴⁹ The unidentate species was shown⁵⁰ to start desorbing from the surface of 85% MgO/CaO oxide only after heating at 300 °C, whereas heating at 500 °C caused its transformation into more stable bridging carbonate by establishing an additional bond between the carbonate oxygen atom and the metal cation. Note that the carbonate region with peaks at 1506 and 1415 cm⁻¹ in FTIR spectra reported⁵⁰ for adsorbed unidentate carbonate species has close resemblance with the carbonate doublet at 1500 and 1402 cm⁻¹ observed in our case after heating the sample to about the same temperature, i.e., 400 °C, for 60–105 min. This provides some evidence that the unidentate carbonate species, initially formed during Mn oxalate thermal decomposition, with increasing time of annealing and formation of MnO as a final product, might partially transform into more stable polydentate structures. Note finally that the carbonate band in the final product is also consistent with the most intense band in the IR spectrum of reference bulk MnCO₃ (Figure 10). Besides, the IR spectrum of MnO produced *in situ* by reduction of MnO₂ in 10% H₂ + He at 400 °C for 1 h displays distinct bands at 976 and 845 cm⁻¹ (Figure 10), which correspond well to the bands at 976 and 839 cm⁻¹ in the IR spectra of the final product.

CONCLUSIONS

This is the first XPS and *in situ* DRIFTS study of isothermal decomposition of MnC₂O₄·2H₂O performed under ultrahigh vacuum and N₂ environmental conditions, respectively. The XP Mn 2p, O 1s, and C 1s core-level spectra, Mn LMV, O KVV Auger spectra, and the infrared absorption bands related to the oxalate ligand, carbonate group, and Mn–O bonding were observed to change with time of annealing to provide information on transient chemical and structural states while decomposing the oxalate structure. After annealing for 25–35 min, we detected the nucleation of two new phases, consistently identified as MnO (XPS, DRIFTS) and MnCO₃ (XPS) or unidentate/polydentate Mn–CO₃ carbonate (DRIFTS). Different from preceding studies, an additional reaction pathway, MnC₂O₄ → MnCO₃ + CO, was identified besides the well-known reaction MnC₂O₄ → MnO + CO + CO₂. With increasing time of annealing the abundance of MnO and MnCO₃ increased along with a decrease in the amount of residual oxalate, with the accumulation of MnCO₃ competing with its thermal decomposition. The concordant time-dependent variations of the satellite-to-peak energy separation and satellite-to-peak intensity ratio in the XP Mn 2p_{3/2} and Mn LMV bands reflected changing ionicity of the –O–Mn–O– bonds upon accumulation of MnO and MnCO₃. The variation of the modified Auger parameters α'(Mn) and α'(O) with time of annealing indicated alterations in the local chemical and structural states of O and Mn atoms during the oxalate decomposition. The kinetics of isothermal decomposition of MnC₂O₄ in vacuum could be well described by the Prout–Tompkins rate equation for an autocatalytic reaction or/and by the Avrami–Erofeev equation with *n* = 4, which corresponds to three-dimensional growth of spherical (or semispherical) nuclei. Both XPS and DRIFTS show that the surface of the final product obtained by annealing the oxalate for 60–150 min was composed of MnO along with MnCO₃ or/and unidentate/polydentate carbonate structures; *ex situ* XRD also evidenced the availability of crystalline MnCO₃ in the decomposed oxalate. Quite interestingly, though not being the main objective of the present study, an appreciable amount of graphitic carbon (up to ~43 atom %)

was found to accumulate during oxalate decomposition at the interface between the MnO particles and the gold foil serving as a sample support.

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Author Contributions

The work was carried out and the manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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